

## Theoretical Exercises – Sheet 1

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1. How is Data Science different from Data Mining? How are they similar?
2. How does Data Mining relate to Machine Learning?
3. Go through the CRISP-DM model and describe in your own words what happens in each phase.
4. Present an example where Data Mining is crucial to the success of a business. What data mining functionalities does this business need? Formulate corresponding business questions.
5. Explain the difference between Association, Segmentation/Clustering, Classification and Regression.
6. Have a look at the efforts in Data Science per phase. Why do you think is data preparation so much effort and why is evaluation not?
7. Provide examples for bad data quality.
8. Provide examples for each attribute type: nominal, ordinal, interval-scaled and ratio-scaled and decide whether it makes sense to calculate mean, median and/or mode.
9. Suppose that the data for analysis includes the attribute *age*. The *age* values for the data tuples are 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70 (in increasing order).
  - a. What is the mean of the data? What is the median?
  - b. What is the mode of the data? Comment on the data's modality.
  - c. Can you find (roughly, by hand) the first quartile  $Q_1$  and the third quartile  $Q_3$  of the data?
  - d. Give the five-number summary of the data and show a boxplot.
10. Suppose that a hospital tested the age and the body fat data for 18 randomly selected adults as follows:

|             |     |      |     |      |      |      |      |      |      |
|-------------|-----|------|-----|------|------|------|------|------|------|
| <i>age</i>  | 23  | 23   | 27  | 27   | 39   | 41   | 47   | 49   | 50   |
| <i>%fat</i> | 9.5 | 26.5 | 7.8 | 17.8 | 31.4 | 25.9 | 27.4 | 27.2 | 31.2 |

|             |      |      |      |      |      |      |      |      |      |
|-------------|------|------|------|------|------|------|------|------|------|
| <i>age</i>  | 52   | 54   | 54   | 56   | 57   | 58   | 58   | 60   | 61   |
| <i>%fat</i> | 34.6 | 42.5 | 28.8 | 33.4 | 30.2 | 34.1 | 32.9 | 41.2 | 35.7 |

- a. Calculate the mean, median and standard deviation of *age* and *%fat*.
- b. Draw a scatter plot based on these two attributes.